

IT INFRASTRUCTURE & SUPPORT DOCUMENTATION

Empowering Communities
Through Technology

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Tech Career Accelerator
Google IT Support Capstone Project

Introduction

Ubuntu Innovations (Pty) Ltd is a growing technology company that requires a secure, scalable, and efficient IT infrastructure to support its daily operations and future business growth. The company consists of 25 employees working across multiple departments including management, finance, human resources, sales and marketing, and software development.

The purpose of this document is to provide a comprehensive overview of the proposed IT infrastructure for the organization. This report includes the company's business requirements, hardware and software recommendations, network topology design, troubleshooting procedures, and security considerations.

The proposed infrastructure has been designed to ensure reliable internet connectivity, secure data storage, centralized network management, and effective technical support procedures for all staff members. The overall solution focuses on improving productivity, maintaining security, and supporting future scalability as the company expands.

Company Overview

Ubuntu Innovations (Pty) Ltd is a small technology-based organization specializing in software development and digital business solutions. The company currently employs 25 staff members who require access to stable computing resources, secure internet connectivity, centralized file storage, and communication systems to support daily business activities.

The organization operates across multiple departments, each requiring different levels of access to hardware, software, and network resources. Employees depend heavily on cloud services, internal file sharing systems, productivity software, and secure wireless connectivity to complete operational tasks efficiently.

Business Operations

The company relies on modern IT infrastructure to support communication, collaboration, software development, administrative operations, and data management. Employees require continuous access to network resources, printers, cloud applications, and secure storage systems.

To support these operations effectively, the organization requires:

- Reliable high-speed internet connectivity
- Secure wired and wireless network infrastructure
- Centralized backup and recovery systems
- Shared printing services
- Protected user authentication and access control
- Scalable hardware and software solutions

Organizational Structure

Ubuntu Innovations (Pty) Ltd consists of multiple departments that require reliable access to the company's IT infrastructure and network resources.

Department	Number of Employees
Executive Management	3
Finance	4
Human Resources	3
Sales & Marketing	7
Software Development	8

Each department requires secure access to company systems, internet services, shared printers, centralized storage, and department-specific software applications. The infrastructure must therefore support both collaboration and controlled access to sensitive company data.

Business Requirements Analysis

Ubuntu Innovations (Pty) Ltd requires a stable, secure, and scalable IT infrastructure capable of supporting daily business operations and future organizational growth.

Internet & Network Access

The organization requires reliable high-speed fiber internet connectivity to support cloud-based systems, internal communications, software development platforms, and general business operations.

The network infrastructure must provide:

- Secure wired and wireless access
- Reliable internal LAN connectivity
- Stable Wi-Fi coverage throughout the office
- Centralized network management
- Support for both desktop and mobile devices

File Storage & Sharing

Employees require centralized storage to securely access, manage, and share company files. A Network-Attached Storage (NAS) device will provide centralized backup capabilities and secure internal file sharing across departments.

The storage solution must support:

- Shared department folders
- Backup and recovery systems
- Secure file permissions
- Reliable data access
- Scalable storage expansion

4.3 Printing Services

The organization requires shared network printing services to support administrative operations, finance documentation, contracts, and internal business records.

Network printers must:

- Remain continuously accessible
- Support multiple departments
- Use static IP configurations
- Integrate with the local network infrastructure

User Accounts & Access Control

Each employee requires a secure user account with role-based access permissions to ensure data confidentiality and system security.

Access control measures include:

- Individual employee login credentials
- Password protection policies
- Department-based permissions
- Restricted access to sensitive information
- Administrative privilege management

4.5 Security Requirements

The company requires strong cybersecurity measures to protect internal systems, company data, and user devices from unauthorized access and cyber threats.

Security requirements include:

- Firewall protection

- Antivirus and anti-malware software
- WPA3 wireless encryption
- Secure password policies
- Software update management
- Secure backup systems

Backup & Recovery

The organization requires a reliable backup and recovery solution to minimize data loss and operational downtime in the event of hardware failure, accidental deletion, or cybersecurity incidents.

Backup procedures should include:

- Automated daily backups
- Centralized NAS backup storage
- Backup verification testing
- Disaster recovery procedures
- Recovery documentation

Hardware Inventory

Hardware Requirements

Ubuntu Innovations (Pty) Ltd requires reliable business-grade hardware to support daily business activities, network infrastructure, and long-term operational growth.

Device	Quantity	Purpose
Desktop Computers	20	Employee workstations
Laptops	5	Mobile productivity
Network Printers	3	Shared office printing
Router & Firewall Appliance	1	Network security and routing
Core Managed Network Switch	1	Central LAN distribution
Wireless Access Points	2	Wireless connectivity
NAS Backup Storage Device	1	Centralized storage and backups
UPS Devices	3	Power protection

Hardware Recommendations

The organization should implement business-grade hardware solutions to ensure reliability, long-term durability, and scalability.

Recommended hardware considerations include:

- Managed network switches for centralized administration
- Firewall-enabled routers for enhanced security
- SSD-based employee workstations for improved performance
- UPS systems to protect critical devices during power outages
- High-capacity NAS storage for secure backup operations

Maintenance Considerations

Regular hardware maintenance is necessary to ensure stable system performance and minimize operational downtime.

Maintenance procedures should include:

- Device inspections
- Hardware cleaning
- Cable management
- Firmware updates
- Component replacement planning
- Equipment lifecycle monitoring

Software Inventory

Standard Company Software

The organization requires standardized software solutions to support productivity, security, communication, and administrative operations.

Software	Purpose
Windows 11 Professional	Operating system
Microsoft Office 365	Productivity suite
Google Chrome	Internet browsing
Microsoft Teams	Communication and collaboration
Antivirus Software	Endpoint security
Adobe Acrobat Reader	PDF document management

6.2 Department-Specific Software

Different departments require specialized software applications to perform operational tasks effectively.

Finance Department:

- Accounting software
- Payroll systems
- Spreadsheet tools

Human Resources:

- Employee management systems
- Recruitment tracking software

Software Development:

- Visual Studio Code
- GitHub
- Development frameworks

- Database management tools

Licensing & Updates

All company software must comply with licensing requirements and receive regular updates to maintain security and functionality.

The organization should implement:

- Licensed software subscriptions
- Automatic update management
- Security patch installation
- Software inventory tracking
- Version control procedures

Network Topology & Infrastructure

Network Architecture Overview

The logical network layout for Ubuntu Innovations (Pty) Ltd implements a centralized hierarchical network design supporting 25 enterprise users.

The infrastructure follows a North-to-South traffic flow model, ensuring that all internal Local Area Network (LAN) resources remain protected from unauthorized public access while maintaining efficient communication between internal systems and external internet services.

The network architecture has been designed to provide:

- Reliable internet connectivity
- Centralized network management
- Secure wireless communication
- High-speed internal data transfers
- Scalable infrastructure support

Network Components

Internet Edge Link

A high-speed fiber connection provides broadband internet access from the Internet Service Provider (ISP) directly into the organization's network infrastructure.

Perimeter Security & Core Routing Layer

The Integrated Edge Router & Firewall serves as the primary security gateway. It manages:

- Network Address Translation (NAT)
- DHCP allocation
- Packet filtering
- Stateful inspection

Core Distribution Layer

A Cat6 Ethernet backbone cable connects the router/firewall to the Core Managed Network Switch, which acts as the central distribution point for all internal network traffic.

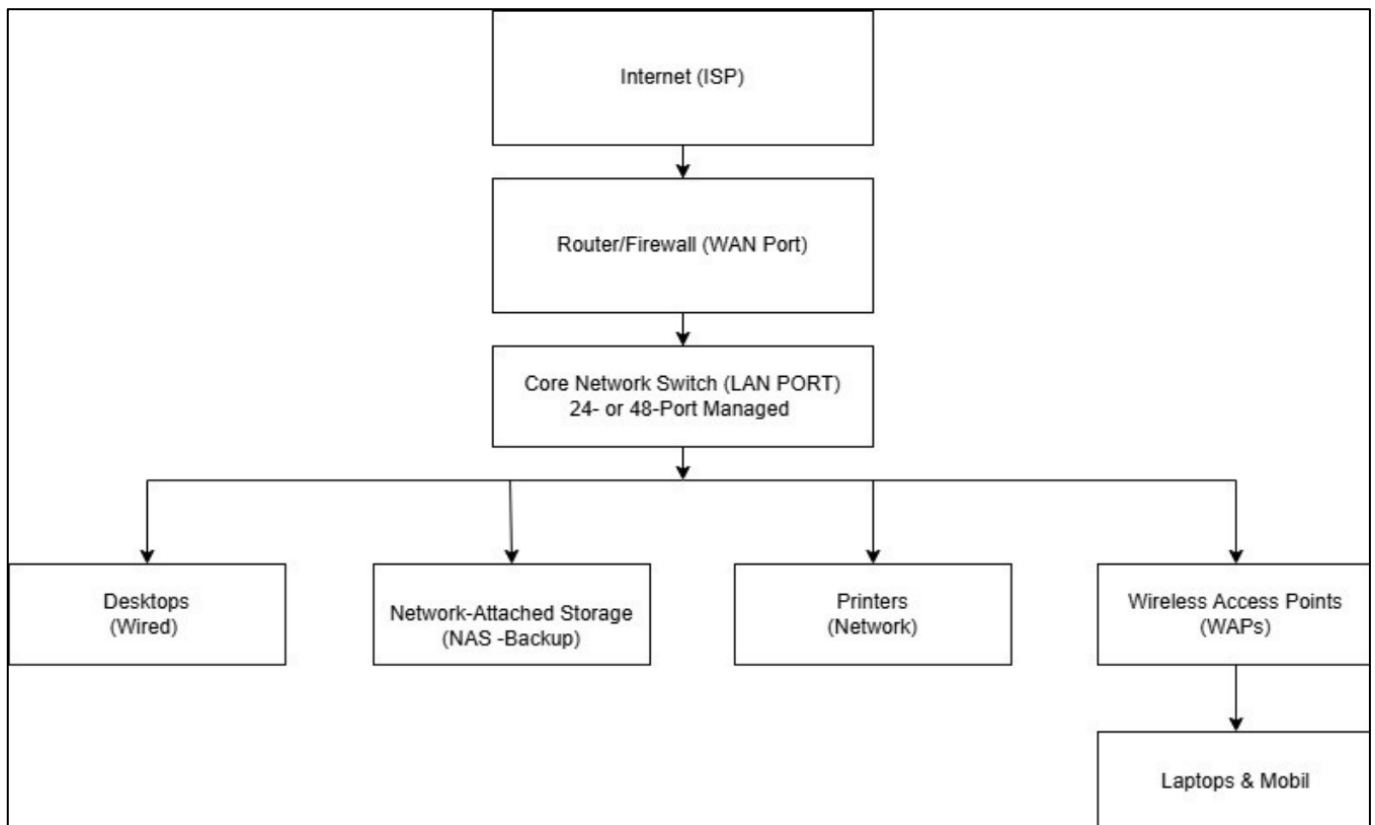
Wired LAN Devices

The managed switch connects directly to:

- Employee desktop workstations
- Shared network printers
- NAS backup storage device

Wireless Infrastructure Layer

Wireless Access Points (WAPs) connect to the managed switch using Power over Ethernet (PoE) connections and broadcast WPA3-secured wireless connectivity for laptops and smartphones.



Enterprise Troubleshooting Guide

Workstation Power Failure

Problem Description

A user reports that their desktop workstation does not power on when the power button is pressed.

Symptoms

- No power indicator lights
- No fan movement
- No POST diagnostic sounds
- Monitor displays "No Signal"

Possible Causes

- Loose power cable
- Faulty wall socket
- Power supply switch turned off
- PSU hardware failure

Troubleshooting Steps

1. Check all power cable connections
2. Test the wall socket using another device
3. Verify the PSU switch position
4. Bypass surge protectors
5. Replace the power cable

Resolution

The technician re-seated the disconnected power cable and restored normal workstation functionality.

Network Layer Drop (No Internet Access)

Problem Description

A workstation can access local network resources but cannot connect to external internet services.

Symptoms

- Websites fail to load
- “No Internet Access” notification
- Internal systems remain accessible

Possible Causes

- Incorrect DHCP configuration
- DNS cache corruption
- Weak wireless signal
- ISP service interruption

Troubleshooting Steps

1. Verify NIC connection indicators
2. Execute `ipconfig /all`
3. Renew DHCP configuration
4. Ping the default gateway
5. Flush the DNS cache

Resolution

The DNS cache was cleared and the workstation's network configuration was refreshed, restoring internet connectivity.

Shared Network Printer Offline

Problem Description

Users are unable to print documents because the shared network printer displays an offline status.

Symptoms

- Print queue errors
- Printer remains inactive
- “Offline” printer status

Possible Causes

- Printer powered off
- Dynamic IP address reassignment
- Loose Ethernet connection
- Print Spooler service failure

Troubleshooting Steps

1. Verify printer power status
2. Check printer network configuration
3. Ping the printer IP address
4. Verify printer port settings
5. Restart the Print Spooler service

Resolution

A static IP reservation was assigned to the printer, restoring proper network printing functionality.

Security & Risk Management

Ubuntu Innovations (Pty) Ltd requires multiple security controls to protect company systems, employee devices, and business data from cybersecurity threats and operational risks.

Security measures include:

- Router firewall protection
- WPA3-secured wireless connectivity
- Antivirus and anti-malware software
- Secure password enforcement
- Role-based access control
- Automated backup procedures
- Software patch management

Potential operational risks include:

- Hardware failure
- Internet outages
- Malware infections
- Unauthorized network access
- Data loss incidents

The organization should implement proactive monitoring, regular backups, and cybersecurity awareness practices to minimize operational disruption.

Conclusion

The proposed IT infrastructure for Ubuntu Innovations (Pty) Ltd provides a secure, scalable, and efficient technology environment capable of supporting current business operations and future organizational growth.

The combination of centralized networking, secure wireless access, business-grade hardware, standardized software deployment, and structured troubleshooting procedures ensures improved operational reliability and reduced downtime.

By implementing the recommendations outlined within this report, the organization will strengthen its cybersecurity posture, improve productivity, and maintain a stable IT environment for all employees.

References

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Ubuntu Documentation

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